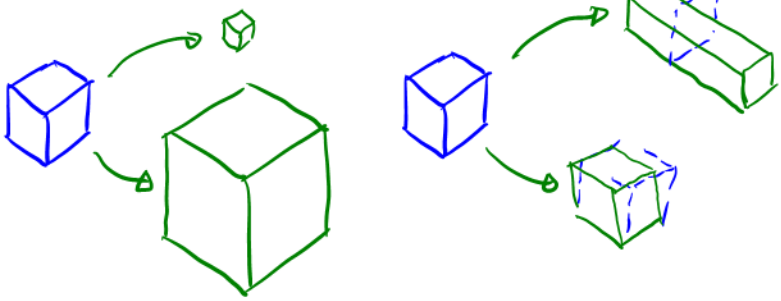
Nonlinear Elasticity

Volumetric – isochoric split

Decomposition: $\underline{\varepsilon} = \frac{1}{3}(\mathrm{tr}\,\underline{\varepsilon})\,\underline{1} + \mathrm{dev}\,\underline{\varepsilon}$

pure volume change — change of shape



Alternative representation:

$$\mathrm{dev}\,\underline{\varepsilon} = \underline{\varepsilon} : \underline{\mathbb{P}} \quad \text{with} \quad \underline{\mathbb{P}} = \underline{\mathbb{I}} - \frac{1}{3}\,\underline{1} \otimes \underline{1}$$

$$\rightsquigarrow \underline{\varepsilon} : \underline{\mathbb{P}} \mathrel{\hat{=}} \varepsilon_{ij}\,\delta_{ik}\,\delta_{jl} - \frac{1}{3}\,\varepsilon_{ij}\,\delta_{ij}\,\delta_{kl}$$

$$= \varepsilon_{kl} - \frac{1}{3}\,\mathrm{tr}\,\underline{\varepsilon}\;\delta_{kl} \mathrel{\hat{=}} \mathrm{dev}\,\underline{\varepsilon} \quad \checkmark$$

$$\rightsquigarrow \mathrm{dev}\,\underline{\varepsilon} : \underline{\mathbb{P}} \mathrel{\hat{=}} \mathrm{dev}\,\varepsilon_{ij}\,\delta_{ik}\,\delta_{jl} - \frac{1}{3}\,\mathrm{dev}\,\varepsilon_{ij}\,\delta_{ij}\,\delta_{kl}$$

$$= \mathrm{dev}\,\underline{\varepsilon}_{kl} - \frac{1}{3}\,\underbrace{\mathrm{tr}(\mathrm{dev}\,\underline{\varepsilon})}_{= \mathrm{dev}\,\underline{\varepsilon} : \underline{1} \mathrel{\hat{=}} (\varepsilon_{kl} - \frac{1}{3}(\mathrm{tr}\,\underline{\varepsilon})\,\delta_{kl})\,\delta_{kl}}\,\delta_{kl}$$

$$= \mathrm{tr}\,\underline{\varepsilon} - \frac{1}{3}(\mathrm{tr}\,\underline{\varepsilon})\,\underbrace{(\mathrm{tr}\,\underline{1})}_{=3}$$

$$\mathrel{\hat{=}} \mathrm{dev}\,\underline{\varepsilon} \qquad = 0$$

Isotropy

Invariants: $\mathcal{J}_1 := \mathrm{tr}(\underline{\varepsilon})$, $\mathcal{J}_2 := \|\mathrm{dev}\,\underline{\varepsilon}\|$

Exple Linear elasticity:

$$\psi(\mathcal{J}_1, \mathcal{J}_2) = \frac{\kappa}{2}(\mathrm{tr}\,\underline{\varepsilon})^2 + \mu \|\mathrm{dev}\,\underline{\varepsilon}\|^2$$

$\Rightarrow$ κ modifies only volumetric part of energy

μ modifies energy in case of shape change

with $\kappa = \frac{E}{3(1-2\nu)}$, $\mu = \frac{E}{2(1+\nu)}$

incompressibility $\rightsquigarrow \nu = 0.5$

$\rightsquigarrow \kappa \to \infty$ ✓

Stresses:

$$\underline{\sigma} = \frac{\partial \psi}{\partial \underline{\varepsilon}} = \frac{\partial \psi}{\partial \mathcal{J}_1} \underbrace{\frac{\partial \mathcal{J}_1}{\partial \underline{\varepsilon}}}_{①} + \frac{\partial \psi}{\partial \mathcal{J}_2} \underbrace{\frac{\partial \mathcal{J}_2}{\partial \mathrm{dev}\,\underline{\varepsilon}}}_{②} : \underbrace{\frac{\partial \mathrm{dev}\,\underline{\varepsilon}}{\partial \underline{\varepsilon}}}_{③}$$

① $\frac{\partial \mathcal{J}_1}{\partial \underline{\varepsilon}} = \underline{1}$

② $$\frac{\partial \mathcal{J}_2}{\partial \mathrm{dev}\,\underline{\varepsilon}} = \frac{\partial \|\mathrm{dev}\,\underline{\varepsilon}\|}{\partial \mathrm{dev}\,\underline{\varepsilon}} = \frac{\partial \left(\mathrm{dev}\varepsilon_{11}^2 + \mathrm{dev}\varepsilon_{12}^2 + \ldots\right)^{\frac{1}{2}}}{\partial \mathrm{dev}\,\underline{\varepsilon}}$$

$$= \frac{1}{2}(\ldots)^{-\frac{1}{2}} \left(2\,\mathrm{dev}\varepsilon_{11} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 2\,\mathrm{dev}\varepsilon_{12} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \ldots \right)$$

$\frac{1}{(\ldots)^{1/2}} = \frac{1}{\|\mathrm{dev}\,\underline{\varepsilon}\|} = \frac{1}{\mathcal{J}_2}$ $\qquad$ $\mathrm{dev}\,\underline{\varepsilon}$

$$= \frac{1}{\mathcal{J}_2}\,\mathrm{dev}\,\underline{\varepsilon}$$

③ $$\frac{\partial\, \mathrm{dev}\underline{\varepsilon}}{\partial \underline{\varepsilon}} = \frac{\partial(\underline{\varepsilon} : \underline{\mathbb{P}})}{\partial \underline{\varepsilon}} = \underline{\mathbb{P}}$$

Insert derivatives

$$\Longrightarrow \underline{\sigma} = \underbrace{\frac{\partial \Psi}{\partial J_1}}\, \underline{1} + \underbrace{\frac{\partial \Psi}{\partial J_2}}\, \frac{\mathrm{dev}\underline{\varepsilon}}{J_2} : \underline{\mathbb{P}}$$

yet to be computed for specific choice of strain energy density

Tangent moduli (instantaneous elasticity tensor):

$$\underline{\mathbb{C}} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} = \frac{\partial \underline{\sigma}}{\partial J_1} \otimes \underline{1} + \frac{\partial \underline{\sigma}}{\partial\, \mathrm{dev}\underline{\varepsilon}} : \underline{\mathbb{P}}$$

Example 1 : Special case linear elasticity

$$\Psi_1 = \frac{1}{2}\varkappa\, \mathcal{J}_1^2 + \mu\, \mathcal{J}_2^2$$

$$\underline{\sigma}_1 = \varkappa\, \mathcal{J}_1\, \underline{1} + 2\mu\, \operatorname{dev} \underline{\varepsilon}$$

$$\underline{\mathbb{C}}_1 = \varkappa\, \underline{1} \otimes \underline{1} + 2\mu\, \underline{\mathbb{P}} \qquad \text{(independent from } \underline{\varepsilon} \text{ !)}$$

Example 2 : Nonlinear isotropic elasticity

Variant (a): $\Psi_2 = \varkappa \frac{1}{3} \mathcal{J}_1^3 + \mu \frac{1}{3} \mathcal{J}_2^3$

$$\rightsquigarrow \underline{\sigma}_2 = \varkappa\, \mathcal{J}_1^2\, \underline{1} + \mu\, \mathcal{J}_2\, \operatorname{dev} \underline{\varepsilon}$$

$$\rightsquigarrow \underline{\mathbb{C}}_2 = 2\varkappa\, \mathcal{J}_1\, \underline{1} \otimes \underline{1} + \mu\, \mathcal{J}_2\, \underline{\mathbb{P}} + \dots$$

$$\dots \mu\, \frac{1}{\mathcal{J}_2}\, \operatorname{dev} \underline{\varepsilon} \otimes \operatorname{dev} \underline{\varepsilon}$$

Problem : $\underline{\varepsilon} = \underline{0} \rightsquigarrow \mathcal{J}_2 = 0 \rightsquigarrow$ Division by zero ↯

Variant (b): $\Psi_2 = \varkappa \frac{1}{3} (\mathcal{J}_1)^3 + \mu \frac{1}{4} \mathcal{J}_2^4$

$$\rightsquigarrow \underline{\sigma}_2 = \varkappa\, \mathcal{J}_1^2\, \underline{1} + \mu\, \mathcal{J}_2^2\, \operatorname{dev} \underline{\varepsilon}$$

$$\rightsquigarrow \underline{\mathbb{C}}_2 = 2\varkappa\, \mathcal{J}_1\, \underline{1} \otimes \underline{1} + \mu\, \mathcal{J}_2^2\, \underline{\mathbb{P}} + 2\mu\, \operatorname{dev} \underline{\varepsilon} \otimes \operatorname{dev} \underline{\varepsilon}$$

Problem: ① $\underline{\underline{\mathcal{E}}} = \underline{\underline{0}} \rightsquigarrow J_1 J_2 = 0$; $\operatorname{dev} \underline{\underline{\mathcal{E}}} = \underline{\underline{0}}$ $\rightsquigarrow \underline{\underline{\mathbb{C}}}_2 = \underline{\underline{\mathbb{O}}}$ $\Rightarrow \underline{\underline{K}} = \underline{\underline{0}}$ ↯

② $\operatorname{tr} \underline{\underline{\mathcal{E}}}$ can become negative

$\rightsquigarrow$ negative value in $\mathbb{C}_{111}$, e.g. if $J_2 = 0$ (purely volumetric deformation)

$\rightsquigarrow \underline{\underline{\mathbb{C}}}$ not positive definite

Variante 3: $\Psi_2 = \Psi_1 + \frac{1}{4} \alpha_{fib} (J_1)^4 + \mu_{fib} \frac{1}{4} J_2^4$

Ψ_1: e.g. lin. elast. matrix material

$\frac{1}{4} \alpha_{fib} (J_1)^4 + \mu_{fib} \frac{1}{4} J_2^4$: e.g. isotropically distributed nonlinear elastic fibers

$$\rightsquigarrow \underline{\underline{\mathcal{S}}}_2 = \underline{\underline{\mathcal{S}}}_1 + \alpha_{fib} J_1^3 \underline{\underline{1}} + \mu_{fib} J_2^2 \operatorname{dev} \underline{\underline{\mathcal{E}}}$$

$$\rightsquigarrow \underline{\underline{\mathbb{C}}}_2 = \underline{\underline{\mathbb{C}}}_1 + 3 \alpha_{fib} J_1^2 \underline{\underline{1}} \otimes \underline{\underline{1}} + \dots$$

$$\dots + \mu_{fib} \left(J_2^2 \underline{\underline{\mathbb{P}}} + 2 \operatorname{dev} \underline{\underline{\mathcal{E}}} \otimes \operatorname{dev} \underline{\underline{\mathcal{E}}} \right) \checkmark$$

$\rightsquigarrow$ dependent on $\underline{\underline{\mathcal{E}}}$

Transverse Isotropy

- Here example of woven fibers in coating
- 2 fiber directions $\underset{\sim}{a}_{warp} \rightsquigarrow \underline{M}_{warp} \rightsquigarrow \mathcal{J}_4^{warp} = tr[\underline{\underline{C}}\,\underline{M}^{warp}]$

$$\underset{\sim}{a}_{fill} \rightsquigarrow \underline{M}_{fill} \rightsquigarrow \mathcal{J}_4^{fill} = tr[\underline{\underline{C}}\,\underline{M}^{fill}]$$

Example 3

$$\Psi_3 = \Psi^{isotr} + \Psi^{warp} + \Psi^{fill}$$

with $\Psi^{isotr} = \Psi_1$

$$\Psi^{warp} = \frac{1}{4}\alpha_{warp}\left(\mathcal{J}_4^{warp}\right)^4$$

$$\Psi^{fill} = \frac{1}{4}\alpha_{fill}\left(\mathcal{J}_4^{fill}\right)^4$$

$$\rightsquigarrow \underline{\underline{S}}_3 = \underline{\underline{S}}_1 + \alpha_{warp}\left(\mathcal{J}_4^{warp}\right)^3 \underline{M}_{warp} + \alpha_{fill}\left(\mathcal{J}_4^{fill}\right)^3 \underline{M}_{fill}$$

$$\rightsquigarrow \underline{\underline{\mathbb{C}}}_3 = \underline{\underline{\mathbb{C}}}_1 + 3\alpha_{warp}\left(\mathcal{J}_4^{warp}\right)^2 \underline{M}_{warp} \otimes \underline{M}_{warp} + 3\alpha_{fill}\left(\mathcal{J}_4^{fill}\right)^2 \underline{M}_{fill} \otimes \underline{M}_{fill}$$